# JIAMING HUANG

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## EDUCATION

**Carnegie Mellon University (GPA=3.99/4.0)** Pittsburgh, PA
*Master of Science in Mechanical Engineering – Research* May 2027
Current Coursework: Advanced Control Systems Integration, Advanced Control for Robotics

**University of Pittsburgh (GPA=3.74/4.0)** Pittsburgh, PA
*Bachelor of Science in Mechanical Engineering, Minor in Computer Science* December 2024
Selected Coursework: Computer Vision, Modern Control Theory, Mechatronics, Embedded System Design

## WORK EXPERIENCE

**Tencent America — Knowhere: Animatronic Robot** Seattle, WA
*Mechanical Engineering Intern* Jun 2026 – Sep 2026

- Integrated VLM-generated dialogue and expression keyframes with TTS and audio-driven lip sync, compensating for audio startup latency to align speech and mouth motion.
- Collected 3,000 motor-babbling samples and trained forward/inverse models; combined model-based optimization with direct interpolation in a hybrid controller for a 24-servo silicone face.
- Mapped webcam-tracked expressions onto the robot using MediaPipe and per-user range calibration; applied EMA smoothing and grouped servo commands to reduce jitter.
- Redesigned a 6-DoF neck as a 3-DoF yaw–pitch–roll mechanism in Fusion 360, reducing actuators from six to three.
- Implemented inverse kinematics with calibrated servo mapping, actuator-limit checks, and motion-path validation before commanding the neck hardware.

**CERLAB - VLM-Guided Visual Inspection with Boston Dynamics Spot** Pittsburgh, PA
*Graduate Research Assistant, Carnegie Mellon University* Aug 2025 – Present

- Built a simulated Spot inspection pipeline (Isaac Sim 5.0, ROS 2, Python, Docker) with RGB-D gauge localization accurate to ~1 cm and automated pan-tilt-zoom framing.
- Compared VLM-only and geometry-verified navigation across 80 simulated episodes; adding a visibility check reduced incorrectly verified readings from 2 to 0 and lowered API cost by 44%.
- Developed a closed-loop VLM viewpoint policy using camera images, an occupancy map, and observation history; improved the viewing angle on the first move in 30/30 trials and recovered the target after occlusion.
- Integrated footprint-aware collision checking and C++ graph-based planning with Spot's walking controller; evaluated performance across obstacle layouts with reproducible experiment records.

## PROJECT EXPERIENCE

**E-Ducati-On: Self-Balancing Motorcycle** Pittsburgh, PA
*Project Member, Carnegie Mellon University* Oct 2025 – Dec 2025

- Implemented PD control for motorcycle roll stabilization using a reaction wheel driven by a BLDC motor.
- Integrated an MPU-6050 IMU over I2C with ESP8266 and the motor controller for attitude feedback and closed-loop motor control at 100 Hz.
- Added velocity desaturation, static-friction compensation, and idle braking to manage reaction-wheel saturation and low-speed motor response.
- Demonstrated static balance, low-speed driving, and recovery from 5–10° roll disturbances on hardware.

**STAIR: Bio-Inspired Walking-to-Rolling Hybrid Quadruped** Pittsburgh, PA
*Project Member, Carnegie Mellon University* Jan 2026 – Apr 2026

- Designed and built an armadillo-inspired quadruped with wheg legs that transitions between walking and rolling.
- Fabricated and soldered a wiring harness integrating an ESP32-S3, Pololu motor drivers, and four DC gearmotors with 64 CPR quadrature encoders.
- Added voltage-divider signal conditioning to level-shift 5 V encoder outputs to 3.3 V ESP32 inputs.
- Developed non-blocking C++ state-machine firmware for walk/retract/roll/deploy transitions, using millis()-based timing and IMU quaternion data for real-time pitch detection.

## SKILLS

Programming & CAD: Python, C++/C, MATLAB/Simulink, Java; SolidWorks, Fusion 360
Controls: PD/PID control, embedded control loops, inverse kinematics, servo calibration
Robotics & Tools: ROS 2, Isaac Sim, Linux/Bash, Docker, Git
Hardware: ESP32/ESP8266, Raspberry Pi; I2C/SPI/UART, IMU integration, signal conditioning, soldering, 3D printing